

GROUP 5

SMART CALORIE AND NUTRITION CALCULATOR USING C++

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I. INTRODUCTION

The Smart Calorie and Nutrition Calculator is designed to help users calculate their daily calorie and nutrition needs. The main problem addressed by the program is the difficulty of understanding daily calorie requirements and calculating them manually. This program will help users to calculate their Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE).

II. BASIC ALGORITHM AND SYSTEM

The program begins by receiving five inputs from the user, Gender, Weight (W) in kilograms, Height (H) in centimeters, Age (A) in years, and Daily Activity Factor Levels. These values are used to calculate the BMR, which represents the calories needed by the body at complete rest.

For Males	For Females
$BMR = 10W + 6.25H - 5A + 5$	$BMR = 10W + 6.25H - 5A - 161$

After getting the BMR, the program calculates TDEE by multiplying the BMR by the activity factor:

$$TDEE = BMR \times \text{Activity Factor}$$

(Little to no exercise: 1.2, Light Activity (1-3 days/week): 1.375, Moderate Activity (3-5 days/week): 1.55, Heavy Activity (6-7 days/week): 1.725, Very Heavy Activity / Athlete: 1.9).

III. MACRONUTRIENT CALCULATION

After finding the TDEE, the program calculates how much protein, carbohydrates, and fat the user should consume. The user chooses the percentage for each nutrient. Protein must be between 10-35%, carbohydrates between 45-65%, and fat between 20-35%. All three percentages must add up to 100%. The formulas used are:

Protein	$(TDEE \times \text{Protein}\% / 4$
Carbohydrates	$(TDEE \times \text{Carbohydrate}\%) / 4$
Fat	$(TDEE \times \text{xFat}\%) / 9$

The numbers 4 and 9 are used because protein and carbohydrates have 4 calories per gram, while fat has 9 calories per gram.

IV. EXAMPLE CALCULATION AND OUTPUT

A Male who weighs 70 kg, is 175 cm tall, is 25 years old, and has a moderate activity level of 1.55. His nutrition target is 20% protein, 55% carbohydrates, and 25% fat.

First, the BMR is calculated:

$$BMR = 10(70) + 6.25(175) - 5(25) + 5 = 1673.75 \text{ kcal/day}$$

Then the TDEE:

$$TDEE = 1673.75 \times 1.55 = 2594.31 \text{ kcal/day}$$

Next, the program calculates the nutrients:

$$\text{Protein} = (2594.31 \times 0.20) / 4 = 129.7 \text{ g}$$

Carbohydrates = $(2594.31 \times 0.55) / 4 = 356.7 \text{ g}$

Fat = $(2594.31 \times 0.25) / 9 = 72.1 \text{ g}$

So, the final daily targets are about 130 g of Protein, 357 g of Carbohydrates, and 72 g of Fat.

V. CONCLUSION

This project demonstrates how C++ can be used to create a practical calculation program. It applies variables, conditions, loops, input validation, formulas, and calculations to solve a real-life problem. The program also makes the calculations faster and easier compared with doing the formulas manually.

VI. REFERENCES

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